

LESSON-1

VECTORS

- i. Addition
- ii. Product
- iii. Cross
- iv. Scalar triple Product
- v. Vector triple Product

Scalar: Magnitude
Vector: Magnitude & Direction

1. Position

2. Null

3. Unit: Magnitude (1)

4. Coplanar $\Rightarrow [abc] = 0$
Vector: (Same plane)

5. Collinear Vectors

Points of Trisection: Two Pt's

Dividing in 1:2 or 2:1

$$OP = xi + yj + zk$$

$$|OP| = \sqrt{x^2 + y^2 + z^2}$$

i. Triangle Law

ii. Parallelogram Law

$$a = a_1i + a_2j + a_3k$$

$$b = b_1i + b_2j + b_3k$$

Collinear if

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$$

$$\vec{a} = k\vec{b}$$

where $k \rightarrow$ constant

Parallel Condition...

6. Non Coplanar

Vector

7. If Three Vectors $\vec{OP}$ $\vec{OQ}$ $\vec{OR}$ are coplanar then

$$[\vec{OP} \ \vec{OQ} \ \vec{OR}] = 0$$

8. Perpendicular ($\theta = 90$) use Dot Product = 0.

9. Parallel ($\theta = 0$)

10. If a, b, c are coplanar (condition)

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0$$

$$\cos \theta = \frac{a \cdot b}{|a||b|}$$

$$\sin \theta = \frac{|a \times b|}{|a||b|}$$

11. $\vec{PQ} = \vec{OP} - \vec{OQ}$ (Positional vector)

12. If TA is Perpendicular to PR, QS

$$\vec{TA} = k \vec{PR} \times \vec{QS}$$

13. (x', y', z') Directional ratios (x, y, z) then Equation of Line is

$$\frac{x-x'}{x_1} = \frac{y-y'}{y_1} = \frac{z-z'}{z_1}$$

14. If $\vec{a}_1, \vec{a}_2$ are collinear then $\vec{a}_2 = \lambda \vec{a}_1$.
Where λ is constant.

Commutative: $\vec{a} + \vec{b} = \vec{b} + \vec{a}$

Associative: $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$

Diagonals: $\frac{1}{2} |\vec{a} \times \vec{b}|$

Sides: $|\vec{A} \times \vec{B}|$

15. Area of Parallelepiped is $[abc]$, Area of Parallelogram:

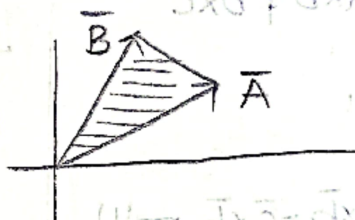
16. Projection of $\vec{a}$ on $\vec{b}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

Projection of $\vec{b}$ on $\vec{a}$ is $\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|}$

17. $i \cdot j = j \cdot k = k \cdot i = 0$

18. $i \cdot i = j \cdot j = k \cdot k = 1$

19. Area of $\triangle OAB = \frac{1}{2} |\vec{A} \times \vec{B}|$



20.

$$A \cdot B = AB \cos \theta$$

$$A \times B = AB \sin \theta$$

$$|\vec{A} \times \vec{B}|^2 + |\vec{A} \cdot \vec{B}|^2 = |\vec{A}|^2 |\vec{B}|^2 \quad (\because \cos^2 \theta + \sin^2 \theta = 1)$$

21. $|\vec{A} + \vec{B}|^2 = |\vec{A}|^2 + |\vec{B}|^2 + 2\vec{A} \cdot \vec{B}$

$$|\vec{A} - \vec{B}|^2 = |\vec{A}|^2 + |\vec{B}|^2 - 2\vec{A} \cdot \vec{B}$$

22.

$$\vec{A} = x_1 \vec{i} + y_1 \vec{j} + z_1 \vec{k}$$

$$\vec{B} = x_2 \vec{i} + y_2 \vec{j} + z_2 \vec{k}$$

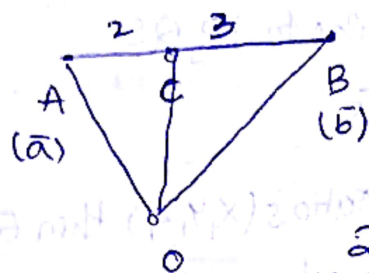
$$\vec{C} = x_3 \vec{i} + y_3 \vec{j} + z_3 \vec{k}$$

$$\vec{AB} = \vec{B} - \vec{A}$$

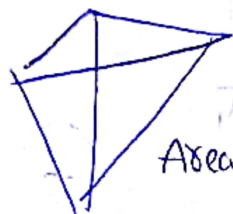
If $\vec{A}, \vec{B}, \vec{C}$ are collinear then

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

23. Section formula:



24.



$$\text{Area} = \frac{1}{2} |\vec{AC} \times \vec{BD}| \text{ (Quadrilateral)}$$

25. $\vec{c} \times (\vec{a} + \vec{b}) = 0$

$\vec{c} \times \vec{d} = 0$

$|\vec{c}| = 0$

$|\vec{d}| = 0$

$\theta = 0^\circ$ Parallel Vectors then $\vec{c} = \lambda \vec{d}$ $\vec{c} = \lambda \vec{a} + \lambda \vec{b}$.

26.

$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c}$

$\vec{a} \times \vec{b} \neq \vec{b} \times \vec{c}$

Pyq's

117Q. $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ — (1)

$\vec{d} \cdot \vec{a} = 24$ — (2)

$\vec{a} = i + 4j + 2k$

$\vec{b} = 3i - 2j + 7k$

$\vec{c} = 2i - j + 4k$

then find $|\vec{d}| = ?$

$\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$

$\vec{d} \times \vec{b} - \vec{c} \times \vec{b} = 0$

$(\vec{d} - \vec{c}) \times \vec{b} = 0$

$\vec{d} - \vec{c} = \lambda \vec{b}$ or $\vec{d} = \lambda(\vec{b} + \vec{c})$

$\vec{d} = \vec{c} + \lambda \vec{b}$

$\vec{d} \cdot \vec{a} = (\vec{c} + \lambda \vec{b}) \cdot \vec{a}$

$\vec{d} \cdot \vec{a} = \vec{c} \cdot \vec{a} + \lambda \vec{b} \cdot \vec{a}$

$24 = 6 + \lambda(9)$ $\lambda = 2$

$\vec{d} = \vec{c} + 2\vec{b}$

$\vec{d} = (2i - j + 4k) + 2(3i - 2j + 7k)$

$\vec{d} = 2i + 6i - j - 4j + 4k + 14k$

$\vec{d} = 8i - 5j + 18k$

$|\vec{d}| = \sqrt{(18)^2 + (5)^2 + 8^2}$

$|\vec{d}| = \sqrt{413}$

27. Obtuse Angle. Take $\vec{a} \cdot \vec{b} < 0$

*** NOTE:

$$|a+b+c| \geq 0$$

$$|a|^2 + |b|^2 + |c|^2 + 2(a \cdot b + b \cdot c + c \cdot a) \geq 0$$

$$3 + 2(a \cdot b + b \cdot c + c \cdot a) \geq 0$$

$$\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{c} + \bar{c} \cdot \bar{a} \geq \frac{-3}{2}$$

— Always holds For

abc of Unit

Vectors ...

$$\therefore |\bar{a} \times \bar{b}|^2 + |\bar{a} \cdot \bar{b}|^2 = |\bar{a}|^2 |\bar{b}|^2$$

line and Point:

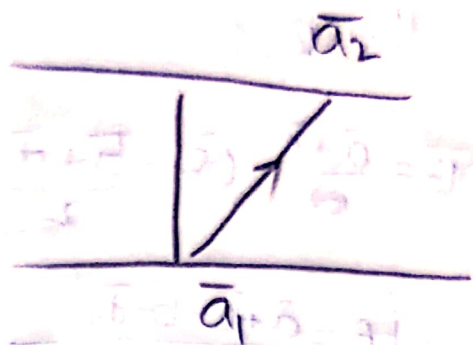
Foot of Perpendicular : Distance Image (C-X)

Case II:

$$r = a_1 + \lambda b_1$$

$$r = a_2 + \mu b_2$$

$$SD = \frac{|a_2 - a_1 \times b|}{|b|}$$



- Equate Given Line to λ to Get General Point on line.

- Line Line :

→ Point of Intersection
→ Shortest Distance

- Intersecting Lines : Shortest Distance is zero.

- Projection of $\vec{b}$ on $\vec{a}$ is $\frac{(\vec{b} \cdot \vec{a}) \vec{a}}{|\vec{a}|^2}$

- Projection of $\vec{a}$ on $\vec{b}$ is $\frac{(\vec{a} \cdot \vec{b}) \vec{b}}{|\vec{b}|^2} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

$$|\vec{a} \times \vec{b} \times \vec{c}| = |\vec{a} \times \vec{b}| |\vec{c}| \cos(\text{Angle b/w } \vec{a} \times \vec{b} \text{ \& } \vec{c})$$

 *** NOTE : For two Given Points Say $P(x_1, y_1, z_1)$ $Q(x_2, y_2, z_2)$
 then Line Equation from $\overline{PQ}$ is $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

$$L: \frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{2}$$

$$\frac{x-x_1}{2} = \frac{y-y_1}{2} = \frac{z-z_1}{2}$$

$$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{2} = \lambda \quad \text{--- (2)}$$

$$\frac{x-\frac{11}{3}}{2} = \frac{y-\frac{11}{3}}{1} = \frac{z-\frac{19}{3}}{2} = \lambda \quad \text{--- (1)}$$

$$x - \frac{11}{3} = 2\lambda \quad x = 2\lambda + \frac{11}{3}$$

$$3\lambda + 11 = 3\lambda + 6$$

$$y - \frac{11}{3} = \lambda \quad y = \lambda + \frac{11}{3}$$

$$\lambda + \frac{11}{3} = \lambda + 2$$

$$z - \frac{19}{3} = 2\lambda \quad z = 2\lambda + \frac{19}{3}$$

$$2\lambda + \frac{11}{3} = \lambda + 1$$

$$x - 1 = \lambda$$

$$x = \lambda + 1$$

$$6\lambda + 11 = 3\lambda + 3$$

$$y - 2 = \lambda$$

$$y = \lambda + 2$$

$$3\lambda + 11 = 3\lambda + 6$$

$$z - 3 = 2\lambda$$

$$z = 2\lambda + 3$$

$$3\lambda = -3$$

$$\boxed{\lambda = -1}$$

- Don't forget Modulus & $\pm$ sign ...

- For $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ $ai+bj+ck$ represents
Vector along Line.

- For Equilateral triangle Orthocentre is Same as Centroid ...

- For Right angled triangle Circumcentre Lies on Hypotenuse (Midpoint)

- G (Centroid) divides Median in $1:2$.

- $a_1a_2 + b_1b_2 + c_1c_2 = 0$ Condition For lines to be Perpendicular ...

- Let $\vec{c}$ be $\perp$ to $\vec{a}, \vec{b}$ then we can write.

$$\vec{c} = \lambda(\vec{a} \times \vec{b})$$

Scalar triple Product:

$$i. \bar{a} \cdot (\bar{b} \times \bar{c}) = [\bar{a} \bar{b} \bar{c}] = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$ii. (\bar{a} \times \bar{b}) \cdot \bar{c} = \bar{a} \cdot (\bar{b} \times \bar{c})$$

$$iii. [\bar{a} \bar{b} \bar{c}] = -[\bar{a} \bar{c} \bar{b}]$$

$$iv. [\bar{a} \bar{b} \bar{c}] = [\bar{b} \bar{c} \bar{a}] = [\bar{c} \bar{a} \bar{b}]$$

$$v. [K_1 \bar{a} K_2 \bar{b} \bar{c}] = K_1 K_2 [\bar{a} \bar{b} \bar{c}]$$

$$vi. [\bar{a} + \bar{b} \bar{c} \bar{d}] = [\bar{a} \bar{c} \bar{d}] + [\bar{b} \bar{c} \bar{d}]$$

vii. Volume of Parallelepiped:

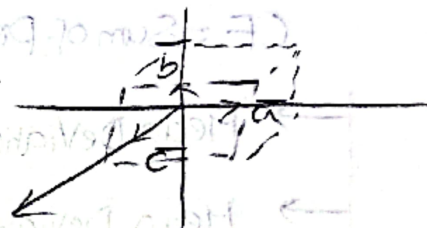
Let $\bar{a}, \bar{b}, \bar{c}$ be coterminus Edges of Parallelepiped

then Volume of Parallelepiped is $[\bar{a} \bar{b} \bar{c}]$

viii. Volume of tetrahedron whose

Coterminus Edges are $\bar{a}, \bar{b}, \bar{c}$

is $\frac{1}{6} [\bar{a} \bar{b} \bar{c}]$



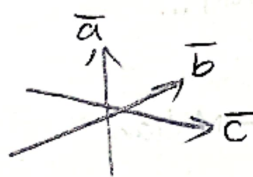
ix: If three Vectors Are coplanar

then $[\bar{a} \bar{b} \bar{c}] = 0$

→ To find one

Unknown Given in Question ...

$$\boxed{\bar{a} = \lambda \bar{b} + \mu \bar{c}}$$



Vector triple product:

$$(i) \bar{a} \times (\bar{b} \times \bar{c}) = \bar{a} \bar{b} \times \bar{c}$$

$$(ii) \bar{a} \times (\bar{b} \times \bar{c}) = (\bar{a} \cdot \bar{c}) \bar{b} - (\bar{a} \cdot \bar{b}) \bar{c}$$

$$(iii) (\bar{a} \times \bar{b}) \times \bar{c} = (\bar{a} \cdot \bar{c}) \bar{b} - (\bar{b} \cdot \bar{c}) \bar{a}$$

Ex: Let Vector $\bar{a}$ Vector $\bar{b}$ lies in the plane

containing $\bar{i} + \bar{j} + \bar{k}$ and $\bar{i} - \bar{k}$ having Magnitude 5 units.

$$\bar{a} = \bar{i} + \bar{j} + \bar{k}$$

$$\bar{b} = \bar{i} + \bar{j} + \bar{k}$$

$$\bar{c} = \bar{i} - \bar{k}$$

$$\bar{a} \cdot \bar{b} = \bar{a} \cdot (\lambda \bar{b} + \mu \bar{c}) = 0$$

$$\text{Method 2: } \bar{d} = \lambda (\bar{a} \times (\bar{b} \times \bar{c}))$$